K-Nearest Neighbors (KNN) for Mechanical Engineering

A comprehensive tutorial on classification and regression for beginners

Prepared as a self-study and teaching resource for Mechanical Engineering students who are new to artificial intelligence and machine learning.

Abstract

This tutorial introduces K-nearest neighbors (KNN) from first principles and explains both KNN classification and KNN regression in a manner intended for Mechanical Engineering students with no prior background in artificial intelligence or machine learning. The tutorial begins with broad ideas such as artificial intelligence, machine learning, supervised learning, classification, regression, and data-driven modelling. It then develops the core intuition behind KNN, namely that nearby observations in a feature space often carry useful information about one another. After that, the discussion moves through geometry, distance measures, formal mathematics, and the effect of the neighbourhood size k on model behaviour. Mechanical Engineering applications, worked examples, and practical guidance are included so that students can connect theory to engineering interpretation. Selected figures are reproduced from academic or reputable educational sources identified in the captions.

Keywords. artificial intelligence, machine learning, supervised learning, K-nearest neighbors, classification, regression, feature space, distance metrics, Mechanical Engineering.

Learning goals

By the end of this tutorial, a beginner should be able to explain what artificial intelligence and machine learning mean, describe where K-nearest neighbors fits within supervised learning, distinguish clearly between classification and regression, interpret the role of distance and neighbourhoods, understand the main KNN equations, and judge whether KNN is a sensible modelling choice for a Mechanical Engineering problem.

Detailed outline of the tutorial

The user requested that the tutorial first be outlined in a way that moves from intuition to geometry to mathematics to Mechanical Engineering applications. The roadmap below therefore appears before the full exposition. Each stage prepares the reader for the next.

Stage	Purpose
1. Mounting bases: artificial intelligence, machine learning, supervised learning	Defines the core ideas needed to place K-nearest neighbors in context. The section introduces data-driven modelling, features, target variables, classification, and regression, and explains why such ideas matter in Mechanical Engineering.
2. Intuition behind K-nearest neighbors	Builds the central idea that similar cases often have similar outcomes. The section uses accessible engineering analogies before any formal mathematics appears.
3. Geometry of neighbourhoods and similarity	Explains how observations become points in a feature space, how distance measures similarity, and why feature scaling matters when variables use different units or magnitudes.
4. KNN classification	Shows how a new point can be assigned to a category by majority vote among nearby labelled observations. Binary and multiclass settings, ties, weighted voting, and the role of k are introduced conceptually.
5. KNN regression	Shows how a new point can be assigned a numerical value by averaging the responses of nearby observations. The section highlights local averaging, weighting, and the difference between predicting classes and predicting quantities.
6. Mathematical mounting bases	Presents the formal notation for the data set, feature vectors, distance measures, neighbour selection, classification rules, regression rules, standardization, and the bias-variance effect of k.
7. Mechanical Engineering applications	Connects KNN to material classification, material strength prediction, machined surface condition assessment, structural health monitoring, bearing failure susceptibility, mechanics of materials and structures risk, coolant quality analysis, and related materials problems.
8. Worked examples	Provides one classification example and one regression example using simplified engineering data, so that the reader can see step by step how neighbours are identified and predictions are formed.

9. Model understanding and practical interpretation	Discusses strengths, limitations, situations where KNN performs well, and conceptual comparisons with other simple modelling approaches.
10. Beginner support and conclusion	Addresses common beginner mistakes, explains training and testing, overfitting, underfitting, and generalization, and concludes with the main lessons that Mechanical Engineering students should retain.

Section summary. The tutorial begins with accessible concepts, then develops geometric intuition, then formal mathematics, and only after that moves to engineering applications and worked examples. This sequence is intended to make a mathematically rigorous topic feel understandable rather than abrupt.

1. Introduction to artificial intelligence and machine learning

Mechanical Engineering increasingly combines traditional physical reasoning with data collected from laboratory tests, test measurements, remote sensing, structural sensors, inspection records, and mechanical systems management systems. When such data are abundant, engineers may use data-driven models to complement mechanistic understanding and empirical design rules. In this setting, artificial intelligence (AI) and machine learning (ML) have become important ideas across mechanics of materials and structures, structural engineering, robotics and vehicle systems engineering, materials engineering, and thermal systems engineering (Baghbani et al., 2022; Gomez-Cabrera & Escamilla-Ambrosio, 2022).

Artificial intelligence (AI) is the broad field concerned with systems that perform tasks associated with intelligent behaviour, such as reasoning, pattern recognition, and decision support. Machine learning (ML) is a subset of AI in which patterns are learned from data rather than being specified entirely in advance by fixed rules (Russell & Norvig, 2021; Bishop, 2006). In many engineering contexts, ML does not replace physical understanding. Instead, it supplements it by finding useful relationships in data when those relationships are too complex to express neatly with a single hand-crafted formula.

A data set is a collection of recorded observations. Each observation is one case, sample, or instance. The measurable quantities used to describe each observation are called features or input variables. The quantity we want to predict is called the target variable, response variable, or output. In supervised learning, each historical observation includes both the input features and the known output. The model then uses these labelled examples to learn how future outputs may be predicted from future inputs (James et al., 2021).

Two of the most important supervised learning tasks are classification and regression. Classification means predicting a category or class. Examples in Mechanical Engineering include material type classification, machined surface condition category, damage state identification, or bearing failure susceptibility class. Regression means predicting a numerical quantity. Examples include mechanical material strength, deflection, deflection, flow rate, or coolant demand. The practical distinction is simple: classification predicts labels, whereas regression predicts numbers.

K-nearest neighbors (KNN) belongs to supervised learning. It is often described as an instance-based, memory-based, or nonparametric method. Instance-based means that predictions are made directly from observed cases. Memory-based means that the reference data set itself remains central to prediction. Nonparametric means that the method does not assume a fixed global algebraic form, such as one straight line or one polynomial, linking the inputs to the output across the entire problem domain (Hastie et al., 2009; Bishop, 2006).

KNN is especially attractive for beginners because its central question is intuitive: when a new case arrives, which previous cases are most similar to it? If engineering behaviour changes gradually with relevant measured properties, nearby historical cases may provide useful guidance. This local perspective is what makes KNN easy to understand conceptually, even though careful mathematical reasoning is still needed for its responsible use.

Foundational terms at a glance

Term	Beginner-friendly meaning
Artificial intelligence (AI)	The broad study of methods that enable systems to perform tasks associated with intelligent behaviour.
Machine learning (ML)	A part of AI concerned with learning useful patterns from data.
Supervised learning	Learning from examples in which both inputs and known outputs are available.
Feature	A measured input variable used to describe one observation.
Target variable	The quantity or label that the model is asked to predict.
Classification	Prediction of a discrete category, such as clay or silt, good or poor, healthy or damaged.
Regression	Prediction of a continuous numerical quantity, such as strength, deflection, or concentration.
Observation	One recorded case in the data set.
Model	A rule or method used to map input information to a prediction.

For Mechanical Engineering students, the relevance of these ideas is practical. Engineering systems are often influenced by many interacting variables, measurement uncertainty, and local mounting conditions. Machine learning methods such as KNN can help summarize past observations and support prediction when similarity between cases is informative. However, their predictions still need engineering judgment, awareness of data quality, and respect for code requirements and physical principles.

Section summary. Artificial intelligence is the broad field, machine learning is a subset of it, and supervised learning uses labelled examples. K-nearest neighbors is a supervised, instance-based, nonparametric method that predicts from nearby similar cases. In Mechanical Engineering, this is relevant whenever measured properties can help identify classes or estimate quantities.

2. Intuition behind K-nearest neighbors

The intuition behind KNN can be stated in everyday language: similar things often behave similarly. Mechanical engineers already reason this way informally. If a new material sample has index properties close to those of previously tested fine-grained materials, an engineer may expect similar classification behaviour. If a new material sample proportion resembles mixes whose strengths are known, those known strengths may help estimate the new strength. KNN turns this common-sense idea into a systematic prediction rule.

The word nearest refers to closeness in terms of measured features. The word neighbors refers to the observed cases that lie closest to the new case of interest. The method does not begin by searching for a single equation that explains the entire data set. Instead, it asks a local question: among the historical cases, which ones are most similar to the current case?

To make this idea precise, three intuitive concepts are needed. First, a neighbourhood is the local group of observations around the new case. Second, similarity means that the feature values are close. Third, distance is the mathematical device used to quantify that closeness. The smaller the distance between two observations, the more similar they are considered to be under the chosen distance measure.

Nearby observations can be informative because many engineering relationships show local continuity. This means that modest changes in inputs often lead to modest changes in outputs, at least within a reasonable range of conditions. For example, small changes in material index properties often indicate related material behaviour, and small changes in material composition may produce related performance. KNN does not prove that such continuity exists, but it exploits it when the data suggest that it does.

The idea should not be interpreted carelessly. Two cases may appear close according to a poor choice of features, and then the resulting prediction may be misleading. For that reason, KNN works best when the selected features are physically meaningful and sufficiently informative. In other words, the method is simple, but the representation of the engineering problem still matters.

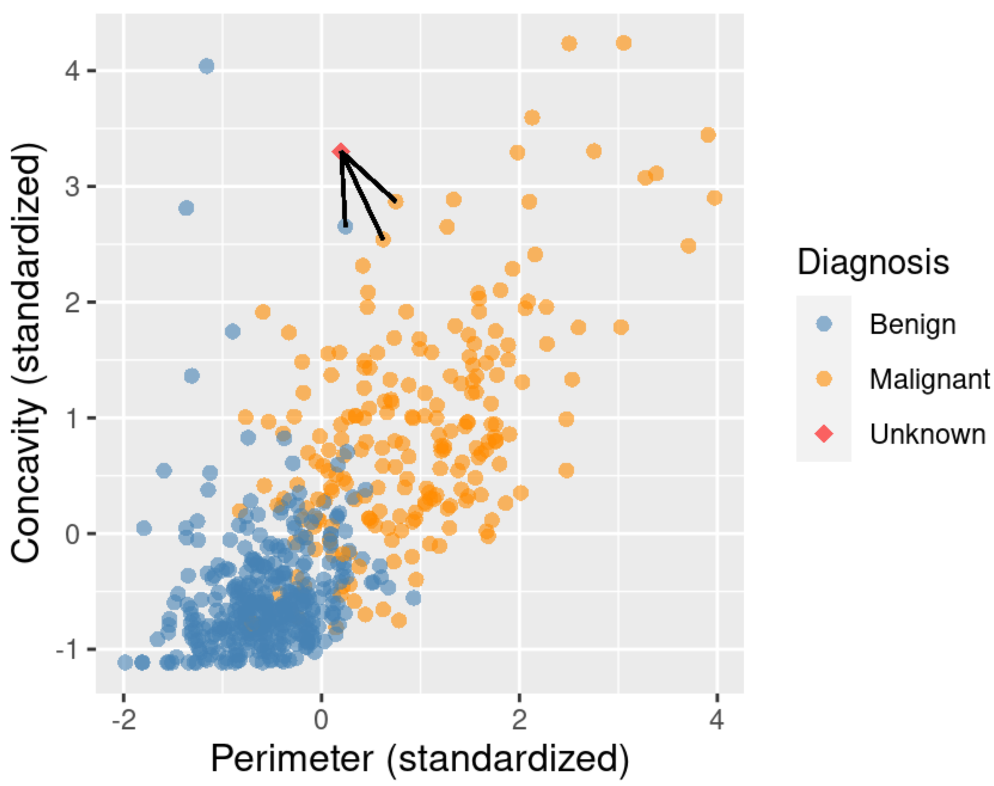

Figure 1. Illustration of KNN classification in a two-feature space using the three nearest neighbors. Although the original example concerns medical data, the geometric principle is identical for Mechanical Engineering variables such as material index properties, sensor features, or materials descriptors. Source: Timbers, Campbell, and Lee (2024), reproduced under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 licence (CC BY-NC-SA 4.0).

Figure 1 helps visualize the central idea. Each observation is represented as a point in a two-feature space. A new point is introduced, and the three nearest observed points are identified. KNN then uses the outputs associated with those neighbours to form a prediction. The figure is useful because it highlights that KNN is fundamentally geometric. Before one can predict, one must decide how points are represented and how closeness is measured (Timbers et al., 2024).

> **Section summary.** KNN rests on the idea that local similarity can be informative. A new case is compared with historical cases, nearby neighbours are identified, and their known outputs guide the prediction. This intuition is simple, but it depends on meaningful features and a sensible definition of distance.

3. Geometry of neighbourhoods, feature space, and distance

The geometry of KNN becomes clear once each observation is viewed as a point in a feature space. A feature space is a coordinate system whose axes are the chosen input variables. If a problem has two features, each observation can be drawn as a point in a pworkcell. If there are three features, the points lie in three-dimensional space. If there are many features, the space has more dimensions than can be drawn easily, but the mathematical idea remains the same.

Suppose a Mechanical Engineering data set records moisture content and dry density for a material sample. Then each sample may be shown as one point whose horizontal position represents moisture content and whose vertical position represents dry density. If the features are instead water-cement ratio, curing age, and fly ash content, then each material sample is a point in a three-dimensional feature space. KNN operates by measuring the closeness of these points.

Distance is therefore a crucial idea. A distance measure, also called a metric, converts differences in feature values into a single nonnegative number. Small distances indicate similarity under the chosen metric, whereas large distances indicate dissimilarity. Different metrics impose different geometric interpretations. A straight-line view of closeness leads to Euclidean distance. A grid-like view leads to Manhattan distance. A broader family that includes both is given by Minkowski distance.

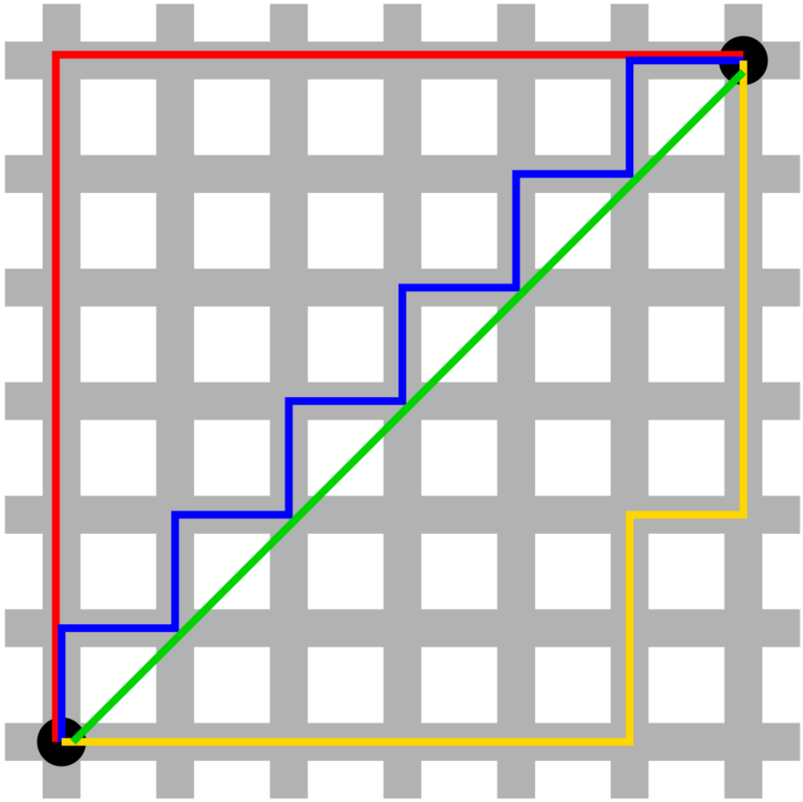

Figure 2. Manhattan and Euclidean distance between the same two points on a grid. The green line shows straight-line Euclidean distance, whereas the coloured broken paths show city-block style production lines associated with Manhattan distance. Source: Psychonaut (2006), Wikimedia Commons, public domain.

The difference shown in Figure 2 matters because a distance measure is not just a technical detail. It expresses what counts as closeness in the problem. Euclidean distance is often natural when straight-line geometric closeness is meaningful. Manhattan distance can be more robust to individual coordinate differences and is sometimes useful when the problem behaves more like movement along coordinate directions. The correct choice depends on the structure of the data and on how the features should be interpreted.

Nearest neighbours are found by computing the distance from the new point to every observed point, sorting those distances from smallest to largest, and keeping the first k observations. If k equals 1, the method uses only the single closest observation. If k is larger, the method uses a broader local neighbourhood.

3.1 Why feature scaling matters

Distance-based methods are sensitive to the magnitudes of the variables. This means that if one feature has a much larger numerical scale than another, it can dominate the distance calculation even if it is not more important from an engineering perspective. For example, in a mechanics of materials and structures problem, depth measured in metres may have larger raw numerical values than water content measured as a fraction. Without scaling, the depth variable may overshadow other meaningful features.

Feature scaling is the process of transforming variables so that their magnitudes become comparable. One common approach is standardization, in which each feature is recentered and rescaled using the mean and standard deviation of that feature in the training data. The purpose is not to change the engineering meaning of the variables, but to ensure that distance reflects relative variation rather than arbitrary unit size (James et al., 2021; Timbers et al., 2024).

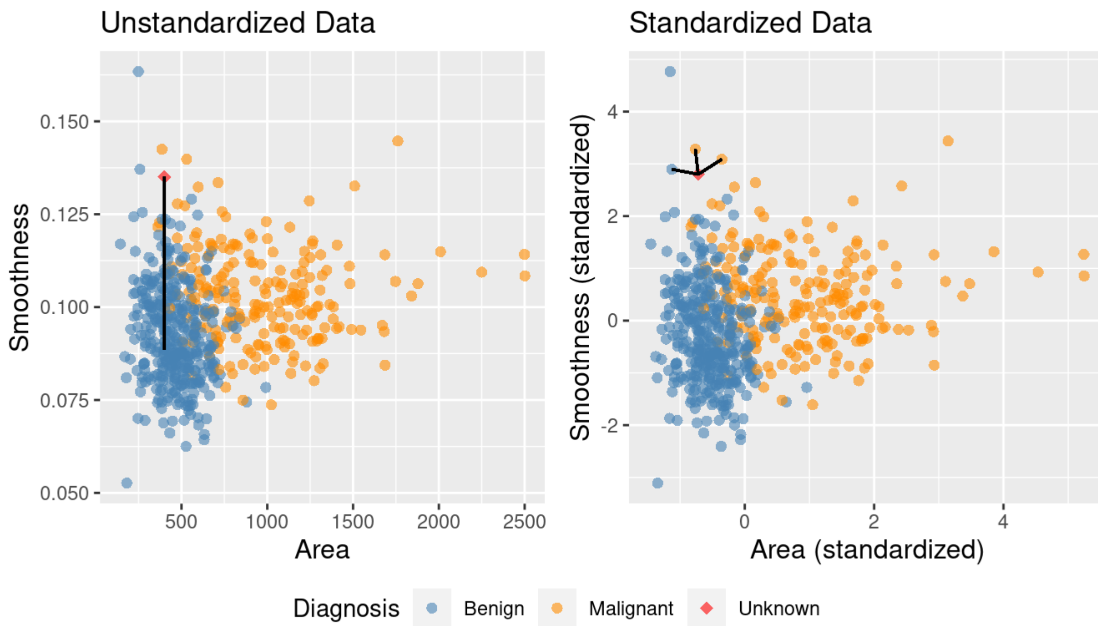

Figure 3. Comparison of nearest-neighbor selection before and after standardization. The example illustrates how an unscaled variable with large numerical magnitude can dominate distance, whereas standardization restores a more balanced notion of similarity. Source: Timbers, Campbell, and Lee (2024), reproduced under CC BY-NC-SA 4.0.

Figure 3 shows that scaling can change which observations count as neighbours. This is one of the most important practical lessons for beginners. In KNN, geometry is not fixed independently of the data representation. It is created by the chosen features, their units, and their scaling. A poor geometric representation produces poor neighbours, and poor neighbours produce poor predictions.

> **Section summary.** In KNN, each observation is a point in a feature space and similarity is measured by distance. Euclidean, Manhattan, and Minkowski distances are common choices. Because distance is sensitive to variable magnitude, feature scaling is often essential in Mechanical Engineering problems where variables have different units or numerical ranges.

4. KNN for classification

KNN classification is used when the target variable is a category. The method assumes that the class labels of nearby observations provide evidence about the class of a new observation. Once the k nearest neighbours have been identified, their labels are examined and combined according to a decision rule.

The most basic rule is majority voting. If most of the k nearest neighbours belong to a particular class, the new point is assigned to that class. In a binary problem, this may mean choosing between two labels such as stable or unstable, healthy or damaged, acceptable or unacceptable. In a multiclass problem, the same idea extends to three or more possible labels such as clay, silt, or sand, or low, moderate, or high risk.

The value of k strongly influences the behaviour of the classifier. A small value of k focuses very tightly on the local neighbourhood. This can make the model sensitive to local structure and able to follow complex decision boundaries. However, it also makes the prediction more sensitive to noise or unusual observations.

A large value of k uses a broader neighbourhood. This usually produces more stable and smoother decisions, but it can also blur important local distinctions.

4.1 Binary and multiclass classification

In binary classification, choosing an odd value of k often reduces the chance of a tie because there are only two classes. Even so, a tie can still arise in some situations, especially when distances are equal or when additional decision criteria are used. In multiclass classification, odd values of k do not guarantee that ties will disappear because votes may still be distributed evenly across several classes.

A tie means that two or more classes receive the same level of support from the selected neighbours. A prediction rule must then specify how to break the tie. Common strategies include choosing the class whose members have the smaller total distance to the query point, using weighted voting so that closer neighbours count more, or applying a predefined rule based on engineering priorities. The important lesson is that ties are not mistakes in the mathematics. They are situations that require an explicit decision rule.

4.2 Weighted KNN classification

Weighted KNN classification assigns greater influence to nearer neighbours and less influence to farther neighbours. This can be useful because not all points within the k-neighbourhood are equally informative. A neighbour that lies extremely close to the new point may deserve a stronger vote than one that lies at the edge of the neighbourhood.

Inverse-distance weighting is a common idea. If the distance is small, the weight is large; if the distance is larger, the weight is smaller. Weighted KNN often improves interpretability in situations where the local structure matters and where there is concern that a few more distant neighbours might distort a simple majority vote. It also provides a natural way to resolve some tie situations.

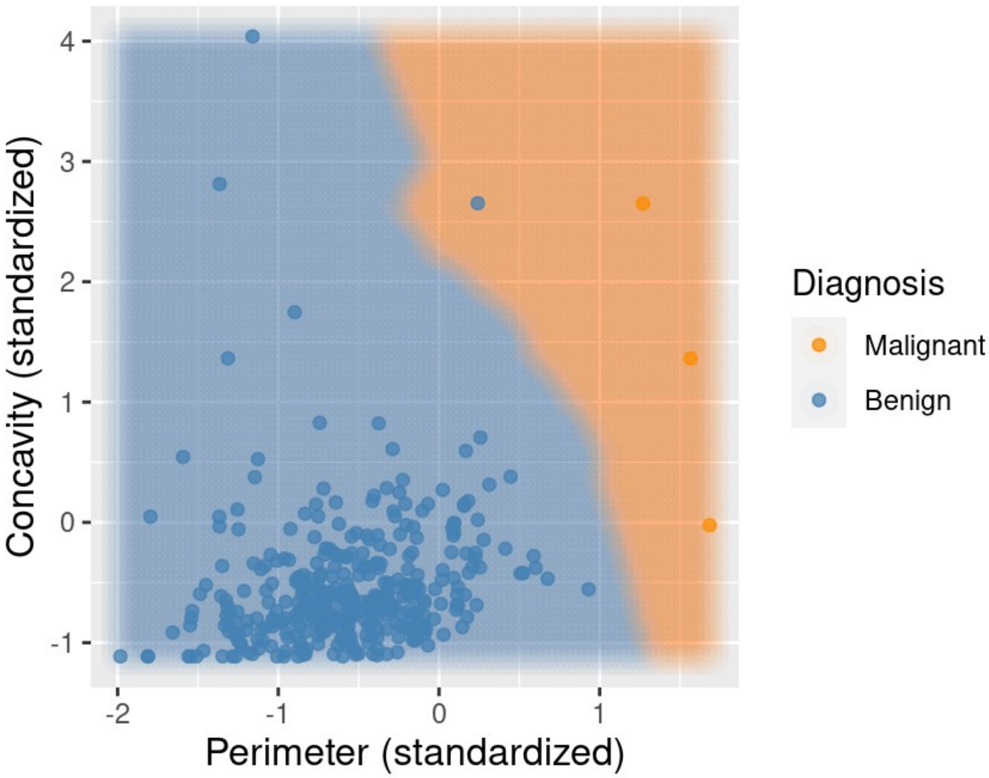

Figure 4. Example of KNN decision regions in a two-feature classification problem. The coloured background shows how local neighbourhoods create a classification boundary, and how a new point is assigned the class dominating its surrounding region. Source: Timbers, Campbell, and Lee (2024), reproduced under CC BY-NC-SA 4.0.

Figure 4 shows that KNN classification can produce decision boundaries that are highly local and irregular. This flexibility is powerful when the true relationship between inputs and classes is not well represented by a simple straight-line boundary. At the same time, it explains why very small k values may overreact to noise. The boundary reflects local geometry, and local geometry becomes unstable if the neighbourhood is too small or the data are noisy.

> **Section summary.** KNN classification predicts categories by examining the labels of nearby observations. Majority voting gives the simplest rule, while weighted voting gives greater influence to closer neighbours. Small k values create flexible but potentially unstable class boundaries, whereas larger k values create smoother and more stable ones.

5. KNN for regression

KNN regression is used when the target variable is numerical rather than categorical. Instead of asking which class is most common among the neighbours, the method asks what numerical value is representative of the neighbours. The usual answer is an average.

In its simplest form, KNN regression predicts the response of a new point by taking the arithmetic mean of the responses of its k nearest neighbours. If the problem is the prediction of compressive strength, the method averages the strengths of nearby historical mixes. If the problem is coolant quality prediction, it averages the observed values from nearby cases in the chosen feature space.

Weighted KNN regression follows the same local logic as weighted classification. Closer neighbours receive larger weights, and the prediction becomes a weighted average. This is often appealing because a point that lies extremely close to the query case may be a more informative guide than points that are only loosely similar.

The effect of local data patterns is especially important in regression. If the target changes smoothly with the inputs, averaging nearby responses may work very well. If the target changes rapidly in a narrow region, a small neighbourhood may capture this local change better than a large one. On the other hand, if the data are noisy, too small a neighbourhood may chase local fluctuations rather than the underlying trend.

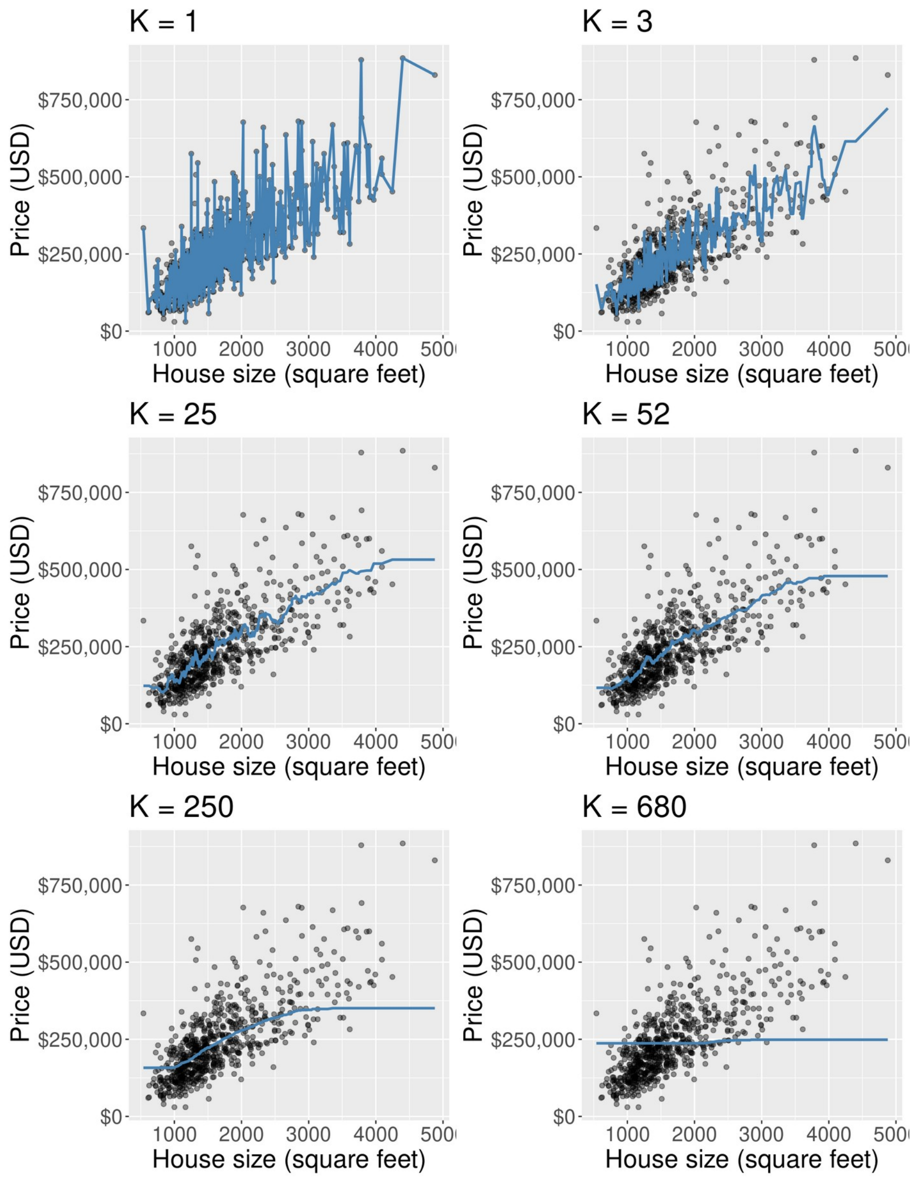

Figure 5. KNN regression predictions for several values of k. Small k values produce highly local fits that can follow noise, whereas larger k values smooth the prediction and may miss local detail. Source: Timbers, Campbell, and Lee (2024), reproduced under CC BY-NC-SA 4.0.

Figure 5 illustrates why KNN regression is often described as a local averaging method. For very small k, the fitted curve can change abruptly from one small region to another. For larger k, the curve becomes smoother because more observations contribute to each prediction. This visual behaviour is closely related to the ideas of overfitting, underfitting, bias, and variance, which will be discussed formally later (Hastie et al., 2009; Timbers et al., 2024).

5.1 Similarities and differences between classification and regression

Aspect	KNN classification and KNN regression
Common feature	Both methods begin by identifying the k nearest neighbours of the new point.
Output type	Classification predicts a category. Regression predicts a numerical value.
How neighbours are combined	Classification uses voting across labels. Regression uses averaging across numbers.
Effect of weighting	In both cases, weighting gives greater influence to closer neighbours.
Engineering interpretation	Both methods rely on the belief that similar engineering cases tend to have similar outcomes.

Section summary. KNN regression predicts numerical values by averaging or weighted averaging among nearby responses. Like KNN classification, it is local and similarity-based. The main difference is the form of the output: classes for classification, numbers for regression.

6. Mathematical mounting bases

The discussion so far has been intuitive and geometric. This section now presents the formal mathematics in a step-by-step way. The notation below follows standard presentations of KNN in introductory and advanced machine learning texts (Bishop, 2006; Hastie et al., 2009; James et al., 2021).

6.1 Data representation

Assume that the training data set contains n observed cases. Each case has m input features and one known output. The full training set may be written as follows.

$$\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$$

Here, n is the number of observed cases. The symbol x in bold denotes the feature vector for one case, and y denotes the known output for that case. In classification, y is a class label. In regression, y is a numerical value.

Each feature vector contains m measured inputs. A typical feature vector may be written as follows.

$$\mathbf{x} = (x_1, x_2, \ldots, x_m)^T$$

The quantity m is the number of features. The superscript T denotes transpose, which indicates that the feature values are viewed as a column vector. In Mechanical Engineering, the entries might represent variables such as moisture content, dry density, water-cement ratio, curing age, crack density, or sensor-derived features.

6.2 Distance measures

To determine which observations are neighbours of a new point x0, a distance must be computed between x0 and every observed point in the training set. Three common distance measures are the Euclidean, Manhattan, and Minkowski distances.

$$d_{\mathrm{E}}(\mathbf{x}_0, \mathbf{x}^{(i)}) = \sqrt{\sum_{j=1}^{m} \left(x_{0j} - x_j^{(i)}\right)^2}$$

Euclidean distance is the straight-line distance between two points in the feature space. The index j runs over the m features.

$$d_{\mathrm{M}}(\mathbf{x}_0, \mathbf{x}^{(i)}) = \sum_{j=1}^{m} \left| x_{0j} - x_j^{(i)} \right|$$

Manhattan distance sums the absolute coordinate-wise differences rather than squaring them. It corresponds to a city-block interpretation of movement in feature space.

$$d_p(\mathbf{x}_0, \mathbf{x}^{(i)}) = \left(\sum_{j=1}^{m} \left| x_{0j} - x_j^{(i)} \right|^p \right)^{1/p}, \quad p \geq 1$$

Minkowski distance is a family of distances indexed by p. When p = 1, Minkowski distance becomes Manhattan distance. When p = 2, it becomes Euclidean distance.

Distance measure	Interpretation
Euclidean distance	Measures straight-line closeness in the feature space. Often used as the default choice.
Manhattan distance	Measures the sum of absolute coordinate-wise differences. Can be less sensitive to a few large coordinate gaps.
Minkowski distance	A flexible family that includes Manhattan and Euclidean distance as special cases.

6.3 Determining the nearest neighbours

After computing distances from the query point x0 to every observation, the distances are ranked from smallest to largest. The index set of the k closest observations is denoted by Nk(x0).

$$\mathcal{N}_k(\mathbf{x}_0) = \{i_1, i_2, \ldots, i_k\}$$

The set Nk(x0) simply records which observed cases are the k nearest neighbours of the query point. Once this set is known, the prediction rule can be applied.

6.4 Classification rule

For classification, one useful way to think mathematically is to estimate, for each class c, the proportion of neighbours that belong to that class. If the indicator function equals 1 when a neighbour has class c and 0 otherwise, then the estimated local class proportion is

$$\widehat{P}(Y=c \mid \mathbf{x}_0)=\frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x}_0)} \mathbf{1}\left(y^{(i)}=c\right)$$

The notation P-hat denotes an estimated quantity. The symbol Y is the class variable, and c denotes one possible class label.

The predicted class is the one with the largest estimated local proportion.

$$\hat{y}(\mathbf{x}_0)=\arg\max_c \widehat{P}(Y=c \mid \mathbf{x}_0)$$

The arg max operation means choose the class c that makes the expression as large as possible.

For weighted KNN, each neighbour receives a positive weight. A common choice is inverse-distance weighting.

$$w_i=\frac{1}{d(\mathbf{x}_0, \mathbf{x}^{(i)})+\varepsilon}$$

Here, epsilon is a small positive constant that prevents division by zero when the query point is identical or extremely close to an observed point.

The weighted classification rule then becomes

$$\hat{y}(\mathbf{x}_0)=\arg\max_c \sum_{i \in \mathcal{N}_k(\mathbf{x}_0)} w_i \mathbf{1}\left(y^{(i)}=c\right)$$

In words, each class receives the sum of the weights of its neighbouring members, and the class with the largest total weighted support is predicted.

6.5 Regression rule

For regression, the prediction is formed from the observed numerical responses of the neighbours. In the unweighted case, the natural local estimate is the arithmetic mean.

$$\hat{y}(\mathbf{x}_0)=\frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x}_0)} y^{(i)}$$

This equation says that the predicted value is the average of the responses of the k nearest neighbours.

In the weighted case, the prediction becomes a weighted average.

$$\hat{y}(\mathbf{x}_0)=\frac{\sum_{i \in \mathcal{N}_k(\mathbf{x}_0)} w_i y^{(i)}}{\sum_{i \in \mathcal{N}_k(\mathbf{x}_0)} w_i}$$

Closer neighbours can therefore contribute more strongly than farther neighbours, while the denominator ensures that the weights sum to one in effect.

6.6 Feature scaling and standardization

Because KNN is distance-based, feature scaling has direct mathematical consequences. A standardization of the jth feature may be written as

$$z_j = \frac{x_j - \mu_j}{s_j}$$

The quantity xj is the original value of the jth feature, mu_j is the mean of that feature in the training data, and s_j is its standard deviation. Standardization gives features a common centre and a common scale so that no variable dominates merely because of its units or raw magnitude.

A crucial practical point is that the scaling parameters must come from the training data. The same transformation is then applied to new points. Otherwise, the geometry of the feature space would change inconsistently between the reference data and the query cases.

6.7 The effect of k on flexibility, bias, and variance

The choice of k controls the size of the local neighbourhood and therefore the flexibility of the model. When k is small, the prediction depends on a very small set of neighbours. This usually gives a flexible model that can adapt to local detail. In statistical language, such a model often has lower bias because it does not smooth the data heavily. However, it often has higher variance because small changes in the data may produce large changes in the prediction.

When k is large, the neighbourhood is wider. The prediction becomes smoother and more stable because many observations contribute. This usually lowers variance but increases bias, because the local prediction may be overly influenced by distant points and may fail to capture genuine local structure. Very small k therefore risks overfitting, which means following noise too closely. Very large k risks underfitting, which means smoothing away important structure (Hastie et al., 2009; James et al., 2021).

Choice of k	Model behaviour	Bias	Variance	Interpretation
Very small k	Very local, highly flexible	Low bias	High variance	Can overfit noise or isolated observations
Moderate k	Balanced local averaging	Moderate bias	Moderate variance	Often the most useful compromise
Very large k	Broad averaging, smoother boundary or curve	High bias	Low variance	Can underfit and miss local structure

6.8 Irrelevant variables and high-dimensional data

KNN works best when distance is meaningful. If many irrelevant variables are included, two genuinely similar engineering cases may appear far apart because of differences in variables that have little relation to the target. This can distort the neighbourhood structure.

A related issue is the curse of dimensionality. This phrase refers to the fact that in high-dimensional spaces, points often become relatively far from one another, and the notion of a tight local neighbourhood becomes less informative. For beginners, the key message is simple: adding more variables is not automatically helpful. Features should be chosen because they carry engineering meaning and likely predictive value.

Section summary. The mathematics of KNN is straightforward once the notation is introduced: represent each observation as a feature vector, choose a distance measure, identify the k nearest neighbours, and combine their outputs by voting or averaging. The choice of k controls flexibility, bias, and variance, while scaling and feature selection determine whether distance is meaningful.

7. Mechanical Engineering applications

KNN has been used across many Mechanical Engineering domains because engineering data often come in the form of measured descriptors paired with known classes or known performance values. The method is most appealing when there is reason to believe that similar measured conditions correspond to similar engineering outcomes, and when a highly flexible local model is preferred over a single global equation.

Application	Task type	Possible inputs	Typical output	Representative source
Material type classification	Classification	Liquid limit, plasticity index, particle-size descriptors, moisture content	Material class or group	Aydin et al. (2023)
Mechanical material strength	Regression	Water-cement ratio, cement content, supplementary cementitious materials, curing age	Strength in megapascal (MPa)	Ghunimat et al. (2023)
Machined surface condition assessment	Classification	Roughness, surface wear, cracking, deflection, production throughput indicators	Condition class or maintenance category	Cano-Ortiz et al. (2022)
Structural health monitoring	Classification	Frequency shifts, damping features, strain or acceleration features	Healthy or damaged state; damage pattern	Vitola et al. (2017); Gomez-Cabrera & Escamilla-Ambrosio (2022)
Bearing failure susceptibility	Classification	Slope, lithology, flow-rate, land cover, distance to production lines or faults	Susceptibility class	Tsangaratos et al. (2015)
Mechanics of Materials and Structures risk identification	Classification	Site descriptors, lubrication and moisture conditions, material strength, excavation context	Risk category	Baghbani et al. (2022)
Coolant quality analysis	Classification or regression	pH, conductivity, particulate concentration, dissolved oxygen, residue variables	Quality class or predicted concentration	Tahraoui et al. (2023)

7.1 Material and mechanics of materials and structures applications

In material classification problems, the goal may be to assign a sample to a material group based on measured properties such as liquid limit, plasticity index, fines content, or particle-size descriptors. Because samples with similar measured properties often occupy nearby regions in a feature space, KNN provides a natural local classification rule. Recent work has examined machine learning methods, including KNN-related approaches, for data-driven material classification tasks (Aydin et al., 2023).

More broadly, mechanics of materials and structures often involves uncertain, heterogeneous, and site-specific data. Reviews of artificial intelligence in mechanics of materials and structures emphasize that data-driven models can assist classification, prediction, and risk assessment when combined with sound mechanics of materials and structures interpretation (Baghbani et al., 2022). KNN is especially useful conceptually because it allows engineers to ask which past sites, samples, or cases most resemble the current one.

7.2 Materials and material applications

Material behaviour depends on many interacting variables, including water content, cement content, supplementary cementitious materials, aggregate properties, admixtures, and curing age. When the relationship between these inputs and compressive strength is complex, a local method such as KNN regression can provide a flexible prediction rule based on similar historical mixes. Studies comparing several machine learning approaches have included KNN regression as one of the candidate methods for material strength prediction (Ghunimat et al., 2023).

7.3 Vehicle systems applications

In machined surface engineering, classification problems arise when segments are assigned to condition categories such as good, fair, or poor, or to maintenance priority classes. Input features may include rut depth, roughness, crack density, falling-weight deflectometer outputs, duty-cycle loading, and environmental indicators. A review of machine learning for machined surface performance monitoring illustrates the growing interest in such data-driven approaches (Cano-Ortiz et al., 2022).

7.4 Structural health monitoring

Structural health monitoring (SHM) refers to the use of measured structural response data to infer the condition of a structure. Features derived from strain, acceleration, modal frequencies, damping, or other sensor signals can be used to distinguish healthy from damaged states. KNN is well suited to this kind of pattern-recognition task because the measured state of a structure can be compared with previously observed states. A sensor data fusion system based on KNN pattern classification has been reported for SHM applications, and recent reviews document the importance of machine learning in this field (Vitola et al., 2017; Gomez-Cabrera & Escamilla-Ambrosio, 2022).

7.5 Bearing failures, risk, and environmental applications

Bearing failure susceptibility assessment often takes the form of classification. The inputs may include topographic, geological, thermal-fluid, land-cover, and mechanical systems-related variables, and the output may be a class such as low, moderate, or high susceptibility. KNN has been explored for such susceptibility mapping because locations that are similar in their controlling variables may share similar hazard levels (Tsangaratos et al., 2015).

Water engineering provides both classification and regression tasks. Coolant quality may be placed into categories, or specific concentrations or residues may be predicted numerically from measured variables. Recent work using KNN in water-related prediction demonstrates the continuing relevance of neighbourhood-based methods in environmental monitoring problems (Tahraoui et al., 2023).

Across all of these application areas, the most important point is not that KNN is universally best. Rather, it is that KNN offers a transparent local baseline: it asks what similar historical engineering cases did, and then uses that local evidence to guide the current prediction.

> **Section summary.** KNN is relevant to Mechanical Engineering because many problems involve classifying or predicting outcomes from measured descriptors of materials, structures, machined surfaces, sites, or environmental conditions. Its local-similarity logic can be useful in material classification, materials prediction, structural health monitoring, hazard assessment, and coolant quality analysis.

8. Worked examples

The following examples are intentionally simplified so that the logic of KNN can be seen without distraction. They are pedagogical examples, not design tools. Their purpose is to show how neighbours are identified and how a final prediction is formed.

8.1 Worked classification example: illustrative material type classification

Suppose a laboratory has a small set of previously labelled fine-material samples. To keep the example focused, assume that only two features are used: standardized liquid limit and standardized plasticity index. Standardized means that the raw variables have already been put on comparable scales. The output is a material class label. Consider the following historical observations.

Sample	Standardized liquid limit	Standardized plasticity index	Known class
S1	1.0	1.2	Clay
S2	1.3	0.9	Clay
S3	0.3	0.7	Silt
S4	0.1	0.5	Silt
S5	-1.0	-1.2	Sand
S6	-1.3	-0.8	Sand

Now consider a new sample with standardized liquid limit 0.8 and standardized plasticity index 0.9. We wish to predict its class using Euclidean distance. The distances from the new sample to each historical sample are shown below.

Sample	Class	Distance setup	Distance
S1	Clay	sqrt((0.8 - 1.0)^2 + (0.9 - 1.2)^2)	0.36
S2	Clay	sqrt((0.8 - 1.3)^2 + (0.9 - 0.9)^2)	0.50
S3	Silt	sqrt((0.8 - 0.3)^2 + (0.9 - 0.7)^2)	0.54
S4	Silt	sqrt((0.8 - 0.1)^2 + (0.9 -	0.81

		0.5)^2)	
S5	Sand	farther away	2.76
S6	Sand	farther away	2.70

The three nearest neighbours are therefore S1, S2, and S3. Their labels are Clay, Clay, and Silt. If k = 3 and simple majority voting is used, the predicted class is Clay because Clay receives two votes and Silt receives one.

If k is increased to 4, the four nearest neighbours are S1, S2, S3, and S4. The class counts then become two Clay and two Silt. This is a tie. The example shows why a tie-breaking rule is needed. One reasonable option is weighted KNN.

Using inverse-distance weights only for these four neighbours gives approximate class supports as follows. Clay receives 1/0.36 + 1/0.50 = 2.78 + 2.00 = 4.78. Silt receives 1/0.54 + 1/0.81 = 1.85 + 1.23 = 3.08. The weighted decision therefore favours Clay. The engineering interpretation is that the new sample lies closer to the known clay samples than to the known silt samples, even though the raw vote count at k = 4 is tied.

This example is deliberately simple, but it captures the main ideas of KNN classification: represent each sample as a point, compute distances, identify neighbours, count or weight their labels, and then interpret the result. In practice, an engineer would also compare such predictions with formal material classification procedures and domain knowledge rather than relying on the data-driven result alone.

8.2 Worked regression example: illustrative mechanical material strength prediction

Suppose that historical material samplees are described only by one feature, the water-cement ratio, and that the target variable is 28-day compressive strength. In reality, material strength depends on many variables, but a one-feature example makes the local averaging idea especially clear.

Mix	Water-cement ratio	28-day compressive strength (MPa)
M1	0.38	48
M2	0.40	46
M3	0.44	42
M4	0.47	39
M5	0.50	36

Now consider a new mix with water-cement ratio 0.43. Using absolute distance in one dimension, the distances to the historical mixes are 0.05, 0.03, 0.01, 0.04, and 0.07, respectively. The three nearest neighbours are therefore M3, M2, and M4, whose strengths are 42, 46, and 39 MPa.

If k = 3 and unweighted KNN regression is used, the predicted compressive strength is the simple average:

Predicted strength = (42 + 46 + 39) / 3 = 42.33 MPa.

If inverse-distance weights are used, then the three weights are approximately 100.00, 33.33, and 25.00. The weighted prediction becomes

Predicted strength = (100×42 + 33.33×46 + 25×39) / (100 + 33.33 + 25) ≈ 42.37 MPa.

The weighted and unweighted predictions are close in this simple example, but the weighted result gives slightly more importance to the closest mix, M3, because its water-cement ratio is only 0.01 away from the new mix. The engineering meaning is straightforward: the new mix is predicted to have a strength close to the strengths of known mixes with similar water-cement ratios.

The example also shows an important limitation. Because the prediction is a local average of observed values, KNN does not extrapolate strongly beyond the range of the data. If the new mix had a water-cement ratio far outside the historical range, the prediction would be much less trustworthy.

> **Section summary.** The worked examples show the full KNN workflow in simple engineering settings. For classification, neighbours vote over labels. For regression, neighbours contribute numerical values that are averaged. Weighted KNN allows closer observations to have stronger influence, which can improve interpretation when some neighbours are much more similar than others.

9. Model understanding and practical interpretation

9.1 Strengths of KNN

- It is intuitive. The method follows the natural idea that similar engineering cases may have similar outcomes.
- It is flexible. Because KNN does not impose a single global formula, it can adapt to nonlinear and irregular local patterns.
- It is versatile. The same general framework handles both classification and regression.
- It is transparent at the case level. An engineer can inspect which historical observations influenced a particular prediction.
- It can serve as a useful baseline. Before adopting more complex methods, KNN often provides a meaningful local comparison point.

9.2 Limitations of KNN

- It is sensitive to feature scaling because distance depends directly on variable magnitudes.
- It is sensitive to irrelevant variables because unhelpful features can distort neighbourhoods.
- It can struggle with class imbalance, where one class dominates the local vote simply because it is much more common.
- It can be affected strongly by noisy observations, especially when k is too small.
- It can become burdensome on large data sets because every new prediction may require many distance calculations.
- It does not extrapolate well beyond the region of observed data.
- It can lose effectiveness in high-dimensional spaces where all points become relatively distant.

9.3 Situations where KNN performs well

KNN often performs well when the following conditions hold: the selected features are meaningfully related to the target, similar cases genuinely tend to have similar outcomes, the data set is reasonably representative of the cases that will later be predicted, the dimensionality is moderate, and there is enough local data density for neighbourhoods to be informative. It can be especially useful when the relationship between inputs and outputs is nonlinear but still locally smooth.

9.4 Challenges that require care

Several challenges deserve particular attention in Mechanical Engineering practice. Feature scaling is essential when variables carry different units. Choice of k matters because it controls the balance between local detail and smoothing. Irrelevant variables can degrade performance by making distance physically meaningless. Class imbalance can bias classification decisions toward common classes. Noisy data can create unstable neighbourhoods. Large data sets can make prediction-time calculations heavy, and sparse high-dimensional data can make close neighbours difficult to identify in a meaningful way.

9.5 Conceptual comparison with other simple modelling approaches

Approach	Core idea	Strength	Limitation
KNN	Local similarity and neighbourhoods	Flexible local predictions; easy to explain conceptually	Sensitive to scaling; weak extrapolation; can be unstable for small k
Linear regression	A global equation relating inputs to a numerical output	Simple global summary and interpretable coefficients	May miss nonlinear local behaviour
Logistic or linear classifiers	A global separating relationship between inputs and class probability	Compact global model; often easier to summarise mathematically	May impose overly simple boundaries
Decision tree	Recursive splitting of the feature space into regions	Easy to visualize as rule paths	Can become unstable and may create abrupt split-based behaviour

The purpose of this comparison is not to rank methods absolutely. Rather, it is to clarify that KNN occupies a distinctive place: it is local rather than global, similarity-based rather than equation-based, and strongly dependent on how distance is defined. These features make it both intuitive and demanding.

Section summary. KNN is attractive because it is intuitive, flexible, and locally interpretable. It also has important limitations, especially sensitivity to scaling, irrelevant variables, noisy data, imbalance, large data sets, and high dimensionality. It is best viewed as a useful local modelling strategy, not as an automatic replacement for engineering judgment or physical reasoning.

10. Beginner support: how to think clearly about KNN

10.1 Common mistakes beginners make

Beginner mistake	Why it is problematic	Better habit
Confusing classification with regression	Misidentifies the modelling task and leads to the wrong form of prediction	Ask first whether the desired output is a category or a numerical value
Ignoring units and scaling	Distance becomes dominated by large-magnitude variables	Place variables on comparable scales before interpreting neighbourhoods
Choosing k arbitrarily	Can lead to overfitting or underfitting	Treat k as a modelling decision that should be justified by unseen data and engineering sense
Using too many irrelevant variables	Meaningful neighbours become harder to identify	Prefer physically meaningful features over indiscriminate feature accumulation

Trusting predictions outside the data range	KNN is poor at extrapolation	Check whether the new case is well represented by historical examples
Assuming nearby means causally related	KNN finds patterns, not physical laws	Interpret results alongside engineering mechanisms and standards

10.2 Training, testing, overfitting, underfitting, and generalization

The training set is the part of the data used as the reference collection from which neighbours are drawn. The testing set is a separate set of observations held aside to evaluate how well the model performs on unseen cases. This distinction matters because a model can appear impressive on the data it already knows while performing poorly on new cases.

Overfitting means that the model follows idiosyncrasies or noise in the training data too closely and therefore does not generalize well. In KNN, overfitting often occurs when k is too small. Underfitting means that the model is too rigid or too smooth and misses important structure. In KNN, underfitting often occurs when k is too large. Generalization is the ability of the model to make useful predictions on new observations drawn from the same type of engineering problem.

10.3 How to decide whether a Mechanical Engineering problem is classification or regression

Question to ask	Model type	Examples
If the desired output is a class label	Classification	Material group, damage state, machined surface category, bearing failure susceptibility class
If the desired output is a numerical quantity	Regression	Compressive strength, deflection, deflection, concentration, discharge

This apparently simple distinction is one of the most important early habits in machine learning. Before choosing a method, state clearly what the model is meant to predict. Ambiguity about the target variable leads to confusion throughout the modelling process.

10.4 Why KNN feels intuitive, and why it still demands care

KNN is often considered intuitive because the prediction rule resembles ordinary reasoning from precedent: look for similar past cases and learn from them. However, intuition alone is not enough. The method still requires careful preparation of the data, sensible feature selection, appropriate scaling, an explicit choice of k, attention to imbalance and noise, and evaluation on unseen cases. In other words, KNN is easy to describe but not effortless to use well.

For Mechanical Engineering students, a productive mindset is the following: begin with the engineering question, identify the target variable, choose input features that carry plausible physical meaning, think carefully about how similarity should be measured, and interpret the resulting prediction in light of codes, mechanisms, and professional judgment. Data-driven modelling is strongest when it remains connected to engineering reasoning.

Section summary. Beginners should focus first on defining the problem clearly, distinguishing classification from regression, understanding training and testing, and recognizing that KNN depends critically on data preparation.

> The method is intuitive because it learns from similar cases, but it remains a serious modelling tool that requires disciplined thinking.

11. Conclusion

K-nearest neighbors is one of the most accessible entry points into machine learning because its central idea is both simple and powerful: use nearby similar cases to guide the prediction for a new case. This tutorial has shown how that idea leads to two related but distinct tasks. In KNN classification, the outputs of nearby neighbours are class labels and the prediction is formed by voting. In KNN regression, the outputs of nearby neighbours are numerical values and the prediction is formed by averaging.

The method becomes rigorous once it is placed in a geometric and mathematical framework. Observations are represented as points in a feature space. Distance measures define similarity. The neighbourhood size k controls the balance between local flexibility and smoothing. Weighted versions allow closer neighbours to matter more. Feature scaling is essential because distance-based methods are sensitive to variable magnitudes. These are not minor technicalities; they are the basisis of the method.

For Mechanical Engineering students, KNN is especially valuable as a conceptual mechanical component between engineering judgement and modern data-driven modelling. It can support tasks such as material classification, material strength prediction, machined surface condition assessment, structural health monitoring, bearing failure susceptibility analysis, mechanics of materials and structures risk identification, and coolant quality prediction. At the same time, it should be used with awareness of its limitations, especially sensitivity to scaling, irrelevant variables, imbalance, noise, and weak extrapolation.

A strong first understanding of KNN gives students more than one algorithm. It provides a way to think about data, similarity, local behaviour, and prediction. That mounting base is useful not only for KNN itself, but also for deeper study of machine learning in Mechanical Engineering.

> **Section summary.** KNN is a local, similarity-based supervised learning method with clear roles in both classification and regression. Its simplicity makes it a powerful teaching tool, while its mathematical structure makes it academically serious. Used thoughtfully, it offers Mechanical Engineering students a strong basis for understanding data-driven prediction.

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